Vahid Danesh

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Education

Stony Brook University (SBU) Stony Brook, NY
PhD in Mechanical Engineering (Minor in Computer Engineering) 2023 – present
- Advisor: Prof. Imin Kao
- Research Topic: AI-Driven Optical Biopsy and Human-Robot Collaborative Resection for Accessible Cancer Care
- GPA: 3.98/4.00

Sharif University of Technology (SUT) Tehran, Iran
BS in Aerospace Engineering 2015 – 2020
- GPA: 3.68/4.00

Selected Projects

Ex-vivo Raman Spectroscopy and AI-Based Classification of Soft Tissue Sarcomas
Led model development and training; built PyTorch ResNet pipeline
- Designed preprocessing, denoising, and classification framework for 286,672 spectra; achieved 97.1% accuracy
- Defined clinical alert metric; 1.46% malignant misclassification rate on 7 patients

Vision-Guided Surgical Navigation for Pelvic Tumor Resections
Validated a vision-guided modular jig system for real-time intraoperative bone resection
- Projects preoperatively planned osteotomy lines directly onto the surgical site
- Published in Journal of Orthopaedic Research; demonstrated sub-millimeter resection accuracy

PandaSim: Physics-Based Dexterous Manipulation with Screw Motion Planning
Manipulation pipeline in Genesis physics-based simulator using Franka Emika Panda robot
- Integrated geometry extraction, screw motion planning, and resolved-rate motion control modules
- Introduced finger-level rotational DoF to enhance dexterity; achieved 96% object reorientation success rate

Publications

Ex-vivo Raman Spectroscopy and AI-Based Classification of Soft Tissue Sarcomas 2025
M Boroji†, **V Danesh**†, D Barrera, E Lee, PG Arauz, RF Farrell, et al., PLoS ONE, 20(9), e0330618

Improved Accuracy in Pelvic Tumor Resections Using a Real-Time Vision-Guided Surgical System 2025
V Danesh, P Arauz, M Boroji, A Zhu, M Cottone, E Gould, FA Khan, I Kao, Journal of Orthopaedic Research

Real-Time, Vision-Guided Orthopedic Surgery for Wide Resection of Primary Bone Sarcomas 2026
G He, **V Danesh**, M Boroji, A Kermanian, P Arauz, FA Khan, I Kao, Handbook of Robotic and Image-Guided Surgery, pp. 633–644, Elsevier

3-Dimensional Kinematics and Kinetics of the Snatch in Elite and Varsity Weightlifters 2025
PG Arauz, G Garcia, J Llerena, M Boroji, **V Danesh**, I Kao, Journal of Biomechanics, 183, 112625

Motion Planning for Object Manipulation by Edge-Rolling 2024
M Boroji†, **V Danesh**†, I Kao, A Fakhari, IEEE/RSJ IROS

Skin Lesion Subtype Classification Using Lesion and Border Radiomic Features 2025
R Basak, **V Danesh**, M Boroji, I Kao, X Qian, T Zhang, et al., AAPM 67th Annual Meeting & Exhibition

Convolutional Neural Network Attention-Guided Segmentation to More Accurately Identify the Edges of Benign and Malignant Skin Tumors 2024
M Boroji†, **V Danesh**†, P Prasanna, J Kim, X Qian, M Khari, et al., AAPM 66th Annual Meeting & Exhibition

Teaching Experience

Teaching Assistant Tehran, Iran
Sharif University of Technology 2017 – 2021
- Dynamics (Fall 2021); Automatic Control (Fall 2019)
- Mechanical Vibration (Spring 2019, Fall 2018, Spring 2018, Fall 2017)
- Thermodynamics I (Spring 2018); Python Programming (Spring 2017)

Leadership and Service

President Stony Brook, NY
Iranian Graduate Student Association, Stony Brook University 2023 – present
- Elected to lead an over 100-member graduate community at Stony Brook University
- Launched peer-support initiatives and cultural programming to improve student well-being

Research Mentor Stony Brook, NY
Stony Brook University 2023 – present
- Mentored 3 undergraduate and 1 high school student on independent research projects
- Projects spanned medical robotics, AI-driven diagnostics, and surgical navigation

Peer Reviewer 2025 – present
IEEE Transactions on Industrial Electronics
- Reviewed manuscripts on robotics, automation, and control systems

Head, Aerospace Eng. Student Council & Academic Affairs Representative Tehran, Iran
Student Council, Sharif University of Technology 2018 – 2019
- Liaised between ~200 students and university leadership on academic policy
- Advocated for curriculum reforms and improved access to academic resources

Honors and Awards

- PhD Works Awards for Career Exploration (ACE) — **Awarded**, Stony Brook University (Spring 2024)
- AIAA Student Engine Design Competition — **1st Place** (2019), American Institute of Aeronautics and Astronautics

Skills

Programming & ML: Python (PyTorch, NumPy, Scikit-learn, OpenCV); MATLAB/Simulink

Tools: Git; Jupyter; LaTeX; Markdown; Linux